

# A High-Power-Density Ultralow-Frequency Energy Harvester Based on a Magnetic Rotor With Built-In Eccentricity and Ferromagnetic Fillers

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**Abstract**—For biomechanical energy harvesters, eccentric-rotor-based structures exhibit great adaptability to limb motions in both frequency and direction, but fail to generate sufficient volume power densities (VPDs) due to volume-inefficient issues. To this end, we propose an internally formed-eccentricity energy harvester (IEEH) with a higher VPD. Its built-in rotor eccentricity realized by filling some fillers made of two different materials between magnets, enlarges the volume proportion of electromagnetic transducing components in a constant device volume, which greatly promotes the magnet-coil interaction. The adopted ferromagnetic fillers also improve its electromagnetic coupling capacity. The impacts of some coil parameters and diverse filler configurations in the rotor are studied for a higher VPD. Based on these mechanisms, the IEEH achieves a superior normalized VPD of  $263.44 \mu\text{W}/(\text{cm}^3 \cdot \text{g}^2)$  at 1.6 Hz, 264.07% higher than that of a conventional counterpart with a close volume. The synergetic effect of the enhanced electromechanical transduction and kinetic responses is analyzed by experiments and fitted-model-based simulations. During on-body tests, the IEEH yields competitive

power and VPD of 3.60 mW and  $140.41 \mu\text{W}/\text{cm}^3$ , revealing its promising future of efficient biomechanical energy harvesting. This work offers a new insight into overcoming the volume-inefficient issues of eccentric-rotor-based energy harvesters undergoing a strict volume limit.

**Index Terms**—Built-in eccentricity, power density, rotational electromagnetic energy harvester (EMEH), ultralow-frequency vibrations.

## I. INTRODUCTION

ENERGY harvesting from omnipresent vibrations like human motions and ocean waves, has attracted broad attention as an encouraging solution to providing renewable power supplies for Internet of Things applications [1], [2]. The electromagnetic energy harvesters (EMEHs), one of the predominant candidates for this technology [3], capture the external vibration energy by their vibration pick-up structures, and then transform it into electricity by the electromagnetic transducing components, i.e., magnets and coils [4]. On this basis, they generate accessible power for electronics in an unattended manner, shedding light on the potential to substitute the conventional batteries requiring repeated charging and replacement [5]. However, pervasively distributed vibrations are mainly featured by ultralow frequencies ( $< 5$  Hz), volatile directions, and the randomness in these two aspects, especially the human limb motions containing adequate biomechanical energy ( $\sim 60$  W) but at 0.5–2 Hz stochastically [6], [7]. Thus, generating sufficient power from such irregular human motions relies heavily on the sophisticated structural design of the EMEHs. Also, the constrained body load-bearing capacity poses a strict limit to their device volumes, making this issue more challenging [8].

Among the reported vibration pick-up structures, the eccentric-rotor-based ones exhibit outstanding adaptability to the above biomechanical features due to their multidirectional effectiveness and easier-to-achieve ultralow working frequencies [4], [9]. A straightforward implement of an eccentric rotor is to build an asymmetrical magnet distribution around a central shaft, where the magnet occupies only part of its movable stroke and sweeps over the coils along a full-cycle trajectory [10], [11]. However, the magnetic flux (MF) change occurs in a limited area

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around the moving magnet, which leaves most coils disengaged from electromechanical transduction and thus deteriorates an EMEH's volume power density (VPD). For example, a low VPD of  $13.85 \mu\text{W}/\text{cm}^3$  was obtained by such a volume-inefficient EMEH ( $171.79 \text{ cm}^3$ ), even though its vibration pick-up was already enhanced by a magnetic-spring mechanism [12]. To fully utilize the coils, Smilek et al. [13] fixed a single coil set at the pivotal point of their EMEH, where the MF changing rate was kept at its maximum throughout the rolling of an eccentric structure configured by magnets and a frame in Tusi couple. The majority (83.3%) of these magnets did not interact with coils, but still increased the overall size. This realization of structural eccentricity also causes inefficient electromagnetic coupling, as the components involved in the electromechanical transduction make up only a small part of the device volume.

Thus, arranging magnets and coil sets identically in a full-circumference way as a rotor-and-stator pair to enlarge their interacting area is more prevalent for eccentric-rotor-based EMEHs, but an additional mass is required for the structural eccentricity [9], [14], [15]. Cai et al. [16] stacked a magnetic ring with an aluminum alloy rotor frame, where a nonmagnetic eccentric mass was attached radially outwards. This externally formed eccentricity by the attached mass restricted the volume proportion of the magnets within a given device volume of  $10.56 \text{ cm}^3$ , and the rotor frame further narrowed the available space for coils to only one side of the magnets. Thus, the output power of this EMEH was confined to  $151 \mu\text{W}$ . Later, Fan et al. [8] designed a self-assembled rotor consisting of transducing magnets and an externally attached magnet for eccentricity without any supporting frame. In this way, the coils were allowed to be deployed on both sides of this magnetic rotor, which contributed to an enhanced power of  $361.39 \mu\text{W}$  at  $3.5 \text{ Hz}$ . But the attached magnet that made bare contribution to power generation still limited the occupied volume proportion of the transducing magnets in the confined harvester size ( $15.9 \text{ cm}^3$ ). Hence, this EMEH design failed to further increase the VPD. Independent semicircular structures were also coaxially assembled with magnets and coils by Hao et al. [15] and Zhang et al. [17], respectively. To achieve ultralow working frequencies, large sizes of these eccentric structures were inevitable [18], [19], which greatly reduced the available space for magnets and coils. Consequently, the lowered volume proportion of the electromechanical transducing components exerts a detrimental impact on the VPDs of such EMEHs, outweighing the power increase by their improved vibration pick-up. These volume-inefficient problems have hindered the VPD enhancement of the existing wearable EMEHs with eccentric vibration pick-up rotors. Therefore, the design of an eccentric rotor that fully utilizes the confined device volume of an EMEH for a higher VPD is still an open issue worth exploring.

To this end, we present an internally formed-eccentricity energy harvester (IEEH) for efficient ultralow-frequency vibration energy harvesting. The eccentric rotor is designed with built-in eccentricity by fillers made of two different materials instead of using an external mass, which enlarges the volume proportion of electromagnetic transducing components within a constant device volume and thus enhances the VPD. Diverse configurations of this eccentric rotor and the coil set thicknesses

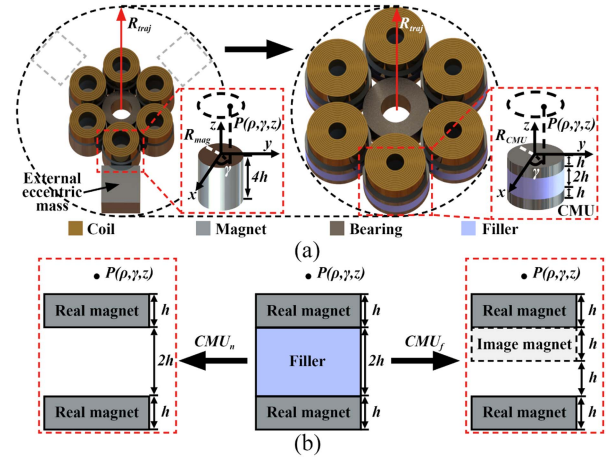


Fig. 1. Concept of the internally-formed-eccentricity magnetic rotor. (a) Comparison between magnetic rotors of the EEEH and the IEEH. (b) Approximated magnet stacks of CMUs.

are studied to further improve the IEEH's performance. Fabricated prototypes are fully characterized in bench tests, and fitted models of their dynamics and power generation are used to better comprehend the measured results. Finally, the results of the human tests involving three subjects demonstrate the improved energy harvesting performance by our approach. Our main contributions are briefly summarized as follows.

- 1) The built-in eccentricity of the magnetic rotor allows an enlarged volume proportion of the electromechanical transducing components in a confined device volume, which greatly promotes the electromechanical transduction of the IEEH.
- 2) The ferromagnetic fillers embedded in the magnetic rotor greatly enhance its electromagnetic coupling capacity (ECC), by increasing the spatial MF changing rate up to its 234.64%.
- 3) Vital factors including the coil set thickness and the ratio of the ferromagnetic ones in all the fillers are thoroughly investigated to further improve the harvester performance.
- 4) Owing to the synergetic effect of dual rotor motion modes and the enhanced electromechanical transduction, the IEEH achieves a superior normalized VPD (NVPD) of up to  $263.44 \mu\text{W}/(\text{cm}^3 \cdot \text{g}^2)$ , 264.07% higher than that of a traditional counterpart with a close volume.

## II. DESIGN AND THEORETICAL FORMULATION

### A. Internally Formed Eccentricity of the Magnetic Rotor

In this work, the structural eccentricity of the magnetic rotor is realized by an asymmetric mass distribution of its embedded fillers. Different from a traditional externally formed-eccentricity energy harvester (EEEH) where the available space for the transducing magnets is radially narrowed by the eccentric mass [see Fig. 1(a)], each disk magnet with an original radius of  $R_{\text{mag}}$  in the rotor is replaced by a composite magnet unit (CMU) in our IEEH, comprising two thinner disk magnets with a larger

radius ( $R_{\text{CMU}}$ ) and one embedded filler. These fillers with identical dimensions are partially made of DT4C electromagnetic pure iron, while the others are made of acrylic. This design endows the rotor with an internal eccentricity without any external mass. To maintain a constant harvester size for comparison purposes, the dimensions of the CMUs are prudently selected, keeping the radius of the rotor movement trajectory ( $R_{\text{traj}}$ ) and the rotor thickness ( $4h$ ) fixed as illustrated in Fig. 1(a). In this way, it is easily derived that  $\sqrt{R_{\text{mag}}^2 + (5R_{\text{mag}})^2} = R_{\text{traj}} = 3R_{\text{CMU}}$ , which shows an increase to its 2.89 times in the cross-sectional area covered by the moving magnets. The magnets and coils are thus allowed to interact in a radially extended area, which greatly promotes the electromechanical transduction of our IEEH.

In view of the high relative permeability of DT4C (about 12000 [20]), the fillers made of DT4C are ferromagnetic and capable of amplifying the generated MF to enhance the electromagnetic coupling between magnets and coils. Based on the mirror image theory [21], such a filler can be approximated as a stacked image magnet with the same geometry and magnetism as the top disk magnet when analyzing the magnetic field above the CMU with this ferromagnetic filler ( $\text{CMU}_f$ ) as shown in Fig. 1(b), which causes an MF amplification. Above a disk magnet of the EEEH or a CMU of the IEEH, the effective MF in a circular region with the same radius as its upper surface at  $z$  is given as

$$\varphi_z(z) = \int_0^R \int_0^{2\pi} B_z(\rho, \gamma, z) d\gamma d\rho \quad (1)$$

where  $B_z(\rho, \gamma, z)$  is the  $z$ -axis MF density (MFD) at arbitrary point  $P(\rho, \gamma, z)$  in this region as seen in Fig. 1(a), and the radius of the integral region follows:

$$R = \begin{cases} R_{\text{mag}}, & \text{EEEH} \\ R_{\text{CMU}}, & \text{IEEH.} \end{cases} \quad (2)$$

To reduce computational costs,  $B_z(\rho, \gamma, z)$  in each case is directly generated by a commercial finite element method (FEM) software (COMSOL Multiphysics). In Fig. 2(a), both a CMU with a nonmagnetic filler ( $\text{CMU}_n$ ) and a  $\text{CMU}_f$  generate obviously higher  $\varphi_z(z)$  than that of a magnet in the EEEH due to the enlarged magnet radius. By comparison, these numerical results also verify the further amplified  $\varphi_z(z)$  of a  $\text{CMU}_f$  by its ferromagnetic filler. To validate the above mirror-image-theory-based analysis,  $\varphi_z(z)$  of the stacked real and image magnets in Fig. 1(b) is presented in Fig. 2(a) using the FEM software and (1), denoted as ‘‘Equivalent stack.’’ A strong consistency with minor deviations is seen between the MF curves of a  $\text{CMU}_f$  and this equivalent magnet stack.

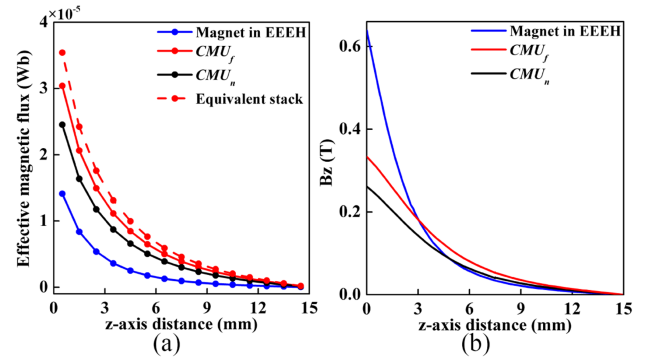


Fig. 2. Decaying magnetic field above the surface of a magnet or a CMU. (a) Effective MF in a circular region. (b) Effective MFD along the central axis.

Likewise, the simulated effective MFD along the central axis of a magnet or a CMU ( $B_z(0, 0, z)$ ) in Fig. 2(b) also reveal the strengthened magnetic field by our design. The enhanced  $B_z(0, 0, z)$  curve of a  $\text{CMU}_f$  is attributed to the ferromagnetic filler acting as an image magnet. According to Fig. 1(b),  $B_z(0, 0, z)$  can be numerically calculated by superimposing the MFDs of these disk magnets, as  $B_z(0, 0, z) = \chi(z)B_r/2$ , where  $B_r$  is the residual MFD of the real magnets and the distance coefficient  $\chi(z)$  is given by [22] and [23], (3) shown at the bottom of this page, where  $R$  is predefined as (2). It is obvious that  $\chi(z)$  of a  $\text{CMU}_f$  is always larger than that of a  $\text{CMU}_n$ , leading to the higher  $B_z(0, 0, z)$ . Besides,  $\chi(z)$  of a magnet in the EEEH decays the most with  $z$  among the three cases due to its smaller  $R$  and causes a rapid decrease of  $B_z(0, 0, z)$ , which even turns lower than the enlarged MFD of a  $\text{CMU}_f$  at a larger  $z$ -axis distance.

The above results and analysis of a CMU have explained the enhancement mechanism of the electromechanical transduction of our proposed rotor. But the configuration of these adopted fillers has dual impacts on both the ultralow-frequency vibration pick-up and electromechanical transduction of the IEEH, which is further elucidated in this section.

## B. Structure Design and Working Mechanism of the IEEH

The architecture of the proposed IEEH is depicted in Fig. 3. It consists of a magnetic rotor comprising six CMUs each with a nonmagnetic filler (acrylic) or a ferromagnetic one (DT4C) embedded between two thin disk magnets (Nd-Fe-B, N52), six self-supported coil sets deployed symmetrically on each side of this rotor, a ball bearing, a central acrylic shaft, and

$$\chi(z) = \begin{cases} \frac{z+4h}{\sqrt{R^2+(z+4h)^2}} - \frac{z}{\sqrt{R^2+z^2}}, & \text{EEEH} \\ \frac{z+2h}{\sqrt{R^2+(z+2h)^2}} - \frac{z}{\sqrt{R^2+z^2}} + \frac{z+4h}{\sqrt{R^2+(z+4h)^2}} - \frac{z+3h}{\sqrt{R^2+(z+3h)^2}}, & \text{CMU}_f \\ \frac{z+h}{\sqrt{R^2+(z+h)^2}} - \frac{z}{\sqrt{R^2+z^2}} + \frac{z+4h}{\sqrt{R^2+(z+4h)^2}} - \frac{z+3h}{\sqrt{R^2+(z+3h)^2}}, & \text{CMU}_n \end{cases} \quad (3)$$

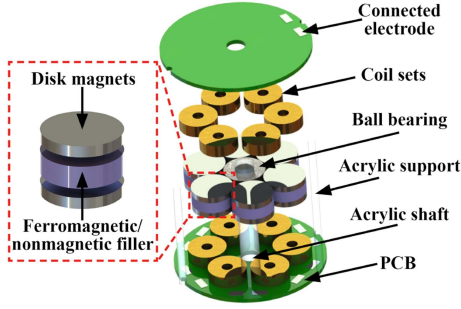


Fig. 3. Architecture of the proposed IEEH.

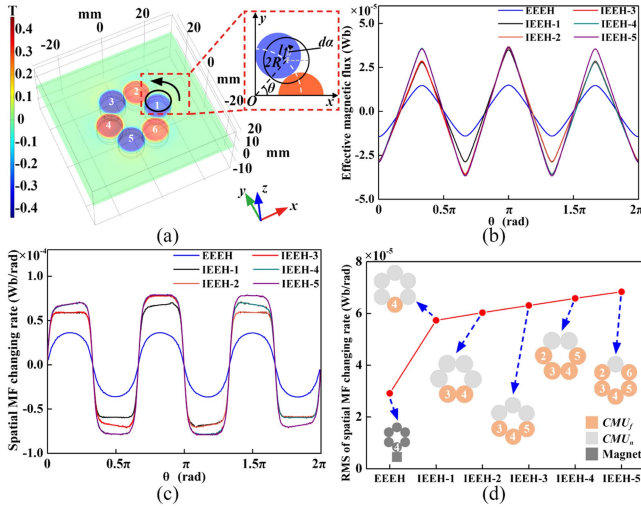


Fig. 4. Generated magnetic fields of the magnetic rotors in the IEEHs and the EEEH. (a)  $B_z$  distribution in the XY-plane 0.5 mm above the upper rotor surface. (b) Spatial distribution of the effective MF. (c) Spatial MF changing rate. (d) RMS value of the spatial MF changing rate.

manufactured acrylic components as structural supports. All the coil sets are connected in series on interconnected printed circuit boards (PCBs) that are firmly glued to the ends of the shaft and also serve as the harvester frame. Based on Faraday's law [24], when the IEEH is driven by external vibrations, an electromotive force (EMF) is generated in these coils due to the moving-rotor-induced MF change, as

$$\varepsilon(t) = -\frac{d\Phi_e(t)}{dt} = -\frac{d\Phi_e(\theta(t))}{d\theta(t)}\dot{\theta}(t) \quad (4)$$

where  $\theta(t)$  and  $\dot{\theta}(t)$  denote the instantaneous relative angular displacement and velocity between the magnetic rotor and coil sets, respectively, and  $\Phi_e(\theta(t))$  is the corresponding effective MF. Due to the coaxially stacked electromagnetic transducing components, only the  $z$ -component of the MF contributes to the EMF, i.e.,  $\Phi_e(\theta(t)) = \Phi_z(\theta(t))$ . According to (4), the spatial MF changing rate ( $d\Phi_e(\theta(t))/d\theta(t)$ ) has a decisive impact on the EMF. Thus, the disk magnets are arranged with alternating polarities to generate significantly changing MFDs on both sides of the magnetic rotor as shown in Fig. 4(a), to promote the electromechanical transduction of the IEEH [3]. The  $z$ -axis distance between the observation plane and the rotor's upper surface in Fig. 4(a),  $z_g$ , is 0.5 mm.

In this work, we define the IEEH with a magnetic rotor configuration of  $n$   $\text{CMU}_f$  and  $6 - n$   $\text{CMU}_n$  as IEEH- $n$  ( $n = 1, 2, \dots, 5$ ), and the CMUs are numbered from 1 to 6 as marked in Fig. 4(a) for convenience in rotor configuration analysis. In each IEEH- $n$ ,  $\text{CMU}_f$  are arranged centered on the 4th  $\text{CMU}$  as briefly sketched in 4(d). Consequently, the 1st  $\text{CMU}$  in each IEEH- $n$  is always a  $\text{CMU}_n$  for structural eccentricity.

We use the changing MF versus  $\theta$  in a single turn of a coil set as marked in Fig. 4(a) to preliminarily reveal the increased spatial MF changing rate by each rotor configuration. The MF is derived by

$$\varphi_z(\theta) = \int_0^R \int_0^{2\pi} B_z(2R \cos \theta + l \cos \alpha, 2R \sin \theta + l \sin \alpha, z_g) dl d\alpha \quad (5)$$

where  $dl$  and  $d\alpha$  are the integral elements over length and angle in this region as illustrated in Fig. 4(a), respectively, and the radius  $R$  is predefined as (2). The effective MFD, i.e.,  $B_z(x, y, z)$ , used in (5) for each IEEH- $n$  or the EEEH is generated by the aforementioned FEM software for ease of calculation. As shown in Fig. 4(b),  $\varphi_z(\theta)$  of these IEEHs is increased to various degrees by both the enlarged interacting area of the transducing components and  $\text{CMU}_f$ , along with an enhanced spatial MF changing rate ( $d\varphi_z(\theta)/d\theta$ ) compared to that of the EEEH [see Fig. 4(c)]. Specifically, a maximum  $d\varphi_z(\theta)/d\theta$  of  $7.93 \times 10^{-5}$  Wb/rad is generated by the IEEH-5, more than double that of the EEEH (about  $3.64 \times 10^{-5}$  Wb/rad). To roughly evaluate the ECC of these magnetic rotors, the root-mean-square (RMS) values of  $d\varphi_z(\theta)/d\theta$  are calculated from Fig. 4(c). In Fig. 4(d), higher RMS values of  $d\varphi_z(\theta)/d\theta$  are achieved by the magnetic rotors of the IEEHs, with an increasing trend as the number of  $\text{CMU}_f$  grows. An increment of 96.76–134.64% in the RMS spatial MF changing rate is reached by these IEEHs compared with the EEEH, where the IEEH-5 with the most  $\text{CMU}_f$  offers the highest ECC. This superiority of the IEEHs to a traditional counterpart lays a solid foundation for their power generation promotion.

Moreover, to analyze the magnetic field distributions of these rotors in the  $z$ -axis, the RMS  $d\varphi_z(\theta)/d\theta$  in a wider  $z$  range of 0.5–14.5 mm are obtained by the same calculation process. In Fig. 5(a), all the RMS  $d\varphi_z(\theta)/d\theta$  decay along the  $z$ -axis, showing that these magnetic rotors provide a lower ECC for the coils farther away from the rotor surface, and thus makes the thickness of coil sets ( $t_c$ ) a vital factor in the power generation performance of these harvesters. To quantify this capacity within the occupied space, a normalized ECC (NECC) is given in Fig. 5(b) for each rotor configuration, represented as  $(2\nabla\varphi_{\text{RMS}})^2/(\pi R_{\text{traj}}^2 \times (2z + 4h))$  ( $\nabla\varphi_{\text{RMS}}$  is the sum of RMS  $d\varphi_z(\theta)/d\theta$  from 0.5 mm to  $z$ ) [9], [21]. Results show that the IEEHs have a higher NECC than the EEEH at arbitrary  $z$ , along with an even larger value as  $n$  grows. This agrees well with our design purpose of the enlarged magnet radius and the ferromagnetic fillers to enhance the electromagnetic coupling. Besides, the highest NECC of each rotor configuration is achieved when the accumulated spatial MF changing rate from

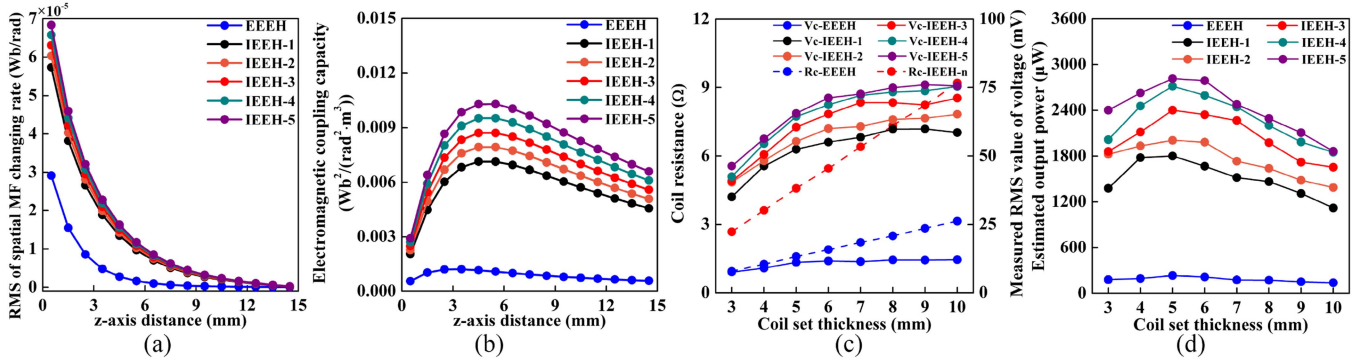


Fig. 5. Analysis of the ECC of magnetic rotors for an optimum coil set thickness in the IEEHs and the EEEH. (a) Decaying RMS spatial MF changing rate versus  $z$ . (b) Normalized ECC. (c) Measured coil resistance and RMS voltage of a single coil set under a constant rotor rotating velocity. (d) Estimated output power of the entire 12 coil sets with a matched load.

0.5 to 4.5–5.5 mm is involved, which indicates that the optimum  $t_c$  probably falls in the range of 4–5 mm in theory.

However, note that this NECC is independent of coil parameters like the inner and outer diameters of the coil sets, and  $t_c$ . These parameters affect both the overall resistance and the utilization of the magnetic fields, which may deviate the optimum  $t_c$  from the theoretical estimation. On this basis, the impact of  $t_c$  on the harvester power generation is studied, within a  $t_c$  range of 3–10 mm. The magnetic rotors of different configurations are driven to rotate at 130 r/min by a motor with a coil set fixed 0.5 mm away from their upper surfaces, respectively. Due to the limit of coil winding fabrication, the inner diameter of a coil set is fixed at 3 mm using 0.2-mm-diameter copper coils, while the outer diameter of a coil set is as close to those of the disk magnets (6 mm for the EEEH) or the CMUs (10 mm for the IEEH- $n$ ) as possible.

The measured resistance ( $R_c$ ) and RMS output voltage ( $V_{c,\text{rms}}$ ) of each coil set are shown in Fig. 5(c). For each case,  $R_c$  grows linearly with  $t_c$ , but the growth rate of  $V_{c,\text{rms}}$  gradually drops before  $V_{c,\text{rms}}$  rises to near its saturation when  $t_c$  reaches 7–8 mm. This result agrees with our analysis of the lower ECC provided by these rotors for the coils at a larger  $z$ -axis distance. Because we use series-connected coil sets of the same dimensions in each IEEH- $n$ , the overall output power with a matched resistive load can be estimated as  $P_{\text{est}} = (12V_{c,\text{rms}})^2 / (4 \times 12R_c)$ . Compared to that of the EEEH in Fig. 5(d),  $P_{\text{est}}$  of each IEEH- $n$  is greatly increased. Further, it is obviously seen that  $P_{\text{est}}$  is also strengthened by more CMU<sub>f</sub> in the IEEHs. Coincidentally,  $P_{\text{est}}$  in each case reaches its maximum when  $t_c = 5$  mm, which determines the optimum  $t_c$  for our harvester prototypes in the next section for their highest accessible output power. It is worth noting that the IEEH- $n$ 's superiority to the EEEH in the VPD is also preliminarily revealed by  $P_{\text{est}}$ , as they can be easily fabricated into the same size due to the same  $t_c$  and an identical occupied space of the moving rotors.

### C. Modeling of the Dynamics and Power Generation

Apart from the electromechanical transduction, the external vibration pick-up by these proposed IEEHs also varies due to their built-in eccentricity by diverse rotor configurations, which

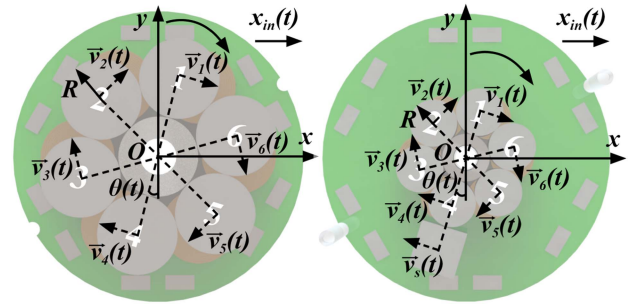


Fig. 6. Schematic diagram of the magnetic rotor movement of the IEEH- $n$  and the EEEH.

finally affects their energy harvesting performance. For each magnetic rotor of the IEEH- $n$  or the EEEH for comparison, when driven by an applied vibration ( $x_{in}(t)$ ), its kinetic energy is

$$T(t) = \sum_{i=1}^6 \frac{1}{2} M_d \|\vec{v}_i(t)\|^2 + \frac{1}{2} M_s \|\vec{v}_s(t)\|^2 \quad (6)$$

where  $\|\vec{v}_i(t)\|$  and  $\|\vec{v}_s(t)\|$  are the magnitudes of velocity vectors of the  $i$ th ( $i = 1, 2, \dots, 6$ ) magnet or CMU in these harvesters, and the attached eccentric mass in the EEEH (see Fig. 6), respectively. The former velocity vector is given as

$$\vec{v}_i(t) = \begin{bmatrix} \dot{x}_i(t) \\ \dot{y}_i(t) \end{bmatrix} = \begin{bmatrix} 2R\dot{\theta}(t) \cos(\theta(t) + \frac{i-1}{3}\pi) + \dot{x}_{in}(t) \\ -2R\dot{\theta}(t) \sin(\theta(t) + \frac{i-1}{3}\pi) \end{bmatrix} \quad (7)$$

where  $R$  is predefined as (2). Likewise,  $\|\vec{v}_s(t)\|^2$  in (6) equals  $16R^2\dot{\theta}^2(t) + \dot{x}_{in}^2(t) - 8R\dot{\theta}(t)x_{in}(t)\cos\theta(t)$ .  $M_d$  denotes the weight of a magnet in the EEEH or a CMU in the IEEH- $n$ , depending on each rotor configuration, as

$$M_d = \begin{cases} m_{d1}, & \text{EEEH} \\ 2m_{d2} + m_f, & \text{IEEH-}n \end{cases} \quad (8)$$

where  $m_{d1}$  and  $m_{d2}$  are the weights of a disk magnet in the EEEH and the IEEH- $n$ , respectively.  $m_f$  is the weight of a nonmagnetic ( $m_{\text{acry}}$ ) or ferromagnetic filler ( $m_{\text{DT4C}}$ ). As for the IEEH- $n$ , the weight of the attached square magnet ( $M_s$ ) is 0 because no external eccentric mass exists. Besides, the potential

**TABLE I**  
PHYSICAL PROPERTIES OF THE FABRICATED PROTOTYPES

| Property                             | EEEH    | IEEH- <i>n</i> |       |       |      |       |
|--------------------------------------|---------|----------------|-------|-------|------|-------|
| <i>n</i>                             | /       | 1              | 2     | 3     | 4    | 5     |
| Magnet material                      | Nd-Fe-B |                |       |       |      |       |
| Residual flux density of magnets (T) | 1.48    |                |       |       |      |       |
| <i>t<sub>c</sub></i> (mm)            | 5       |                |       |       |      |       |
| Inner diameter of a coil set (mm)    | 3       |                |       |       |      |       |
| Outer diameter of a coil set (mm)    | 6       | 10             |       |       |      |       |
| <i>m<sub>d1</sub></i> (g)            | 1.51    | /              |       |       |      |       |
| <i>m<sub>d2</sub></i> (g)            | /       | 1.15           |       |       |      |       |
| <i>m<sub>acry</sub></i> (g)          | /       | 0.26           |       |       |      |       |
| <i>m<sub>DT4C</sub></i> (g)          | /       | 2.37           |       |       |      |       |
| <i>M<sub>s</sub></i> (g)             | 1.92    | /              |       |       |      |       |
| <i>R</i> (mm)                        | 3.175   | 5.55625        |       |       |      |       |
| <i>R<sub>in</sub></i> (Ω)            | 18.4    | 55.1           | 54.5  | 55    | 55   | 55.6  |
| Rotor weight (g)                     | 10.98   | 17.47          | 19.58 | 21.69 | 23.8 | 25.91 |
| Device diameter (cm)                 | 4.00    |                |       |       |      |       |
| Device thickness (cm)                | 1.98    | 2.00           | 2.02  | 2.03  | 2.04 | 2.02  |

energy of the rotor is formulated as

$$V(t) = \sum_{i=1}^6 M_d g y_i(t) + M_s g y_s(t) \quad (9)$$

where  $y_s(t)$  and  $y_i(t)$  are the central  $y$ -positions of the square magnet and the  $i$ th magnet or CMU.  $g$  is the gravitational acceleration ( $g = 9.8 \text{ m/s}^2$ ). The Euler–Lagrange equations are then derived by

$$\begin{cases} \frac{d}{dt} \frac{\partial \mathcal{L}(t)}{\partial \dot{\theta}(t)} - \frac{\partial \mathcal{L}(t)}{\partial \theta(t)} + \frac{\partial R_b(t)}{\partial \dot{\theta}(t)} = 0 \\ \mathcal{L}(t) = T(t) - V(t) \end{cases} \quad (10)$$

where  $R_b(t)$  is the Rayleigh dissipation function ( $R_b(t) = \frac{1}{2} b \dot{\theta}^2(t)$ ) with  $b$  containing the mechanical and electrical damping coefficients ( $b_m$  and  $b_e$ ) [19], [25]. Based on (6)–(10), the motion governing equation of each magnetic rotor can be obtained as follows [8], taking the IEEH-3 and the EEEH as examples

$$\ddot{\theta}(t) = \begin{cases} \frac{4[\ddot{x}_{in}(t) \cos \theta(t) - g \sin \theta(t)] M_s R - b \dot{\theta}(t)}{(24m_{d1} + 16M_s) R^2}, & \text{EEEH} \\ \frac{4[\ddot{x}_{in}(t) \cos \theta(t) - g \sin \theta(t)] (m_{DT4C} - m_{acry}) R - b \dot{\theta}(t)}{12(4m_{d2} + m_{DT4C} + m_{acry}) R^2}, & \text{IEEH-3.} \end{cases} \quad (11)$$

According to (11), the IEEH-3 and the EEEH are likely to exhibit similar nonlinear dynamics under applied vibrations as indicated by their homogeneous rotor motion models. But differences in the physical properties and coefficients will lead to diverse behaviors in their vibration pick-up performance like  $\theta(t)$ ,  $\dot{\theta}(t)$ , and the frequency response. For example, distinct natural frequencies of the rotors can be derived as 2.87 and 1.97 Hz for the EEEH and the IEEH-3, respectively, assuming a small initial displacement without any input based on the parameters in Table I and the above models [13].

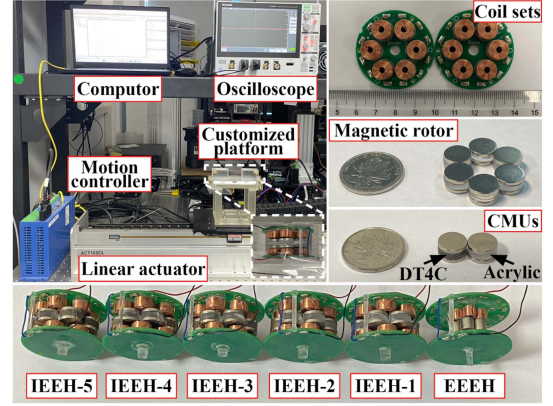


Fig. 7. Fabricated prototypes and the experimental setup.

The steady-state output power of each harvester with a matched resistive load ( $P_{ss}$ ) can be formulated by the electrical damping in our built models as

$$P_{ss} = \frac{b_e}{\tau} \int_{t_0}^{t_0+\tau} \dot{\theta}^2(t) dt = b_e \dot{\theta}_{rms}^2 \quad (12)$$

where  $\tau$  is the steady-state excitation period and  $\dot{\theta}_{rms}$  is the RMS value of  $\dot{\theta}(t)$ . Since  $P_{ss}$  is defined by  $P_{ss} = \varepsilon_{rms}^2 / (4R_{in})$  ( $R_{in}$  denotes the internal resistance of each harvester), the relation between the RMS value of the EMF and  $\dot{\theta}_{rms}$  is easily obtained as  $\varepsilon_{rms} = 2\sqrt{b_e R_{in}} \dot{\theta}_{rms}$ . On this basis, accurate rotor motion models can be fitted by the measured RMS open-circuit voltages ( $V_{oc,rms}$ ), and thus the dynamics of different magnetic rotors underlying the power generation results of these harvesters can be further investigated by simulations of  $\theta(t)$  and  $\dot{\theta}(t)$ . The model fitting process using the same input  $x_{in}(t)$  as those applied in bench tests is described in detail in Section III-C.

### III. EXPERIMENTAL STUDIES AND DISCUSSIONS

#### A. Prototype and Experimental Setup

Prototypes of the proposed IEEH- $n$  and the EEEH for comparison are fabricated with designed properties listed in Table I to demonstrate the proof of concept. As shown in Fig. 7, DT4C electromagnetic pure irons and the acrylic with the same radius as that of the magnets are exploited, respectively, as ferromagnetic and nonmagnetic fillers in the CMUs. Six coil sets on each side of every magnetic rotor are fabricated with the optimum  $t_c$  of 5 mm. Owing to the consistent dimensions of the main components, these prototypes have similar device volumes around  $25 \text{ cm}^3$ .

Repeated sinusoidal reciprocating motions are provided by a programmed linear actuator (Aerotech ACT165DL) and a motion controller (Aerotech Soloist HPe 10) as seen in Fig. 7, with excitation accelerations ranging from 0.1 to 1.5 g at 1 to 10 Hz. The prototypes are fixed on a customized platform upon its movable stage to avoid electromagnetic interference, of which the open-circuit voltages ( $V_{oc}$ ) are measured by an oscilloscope (Tektronix MDO34).

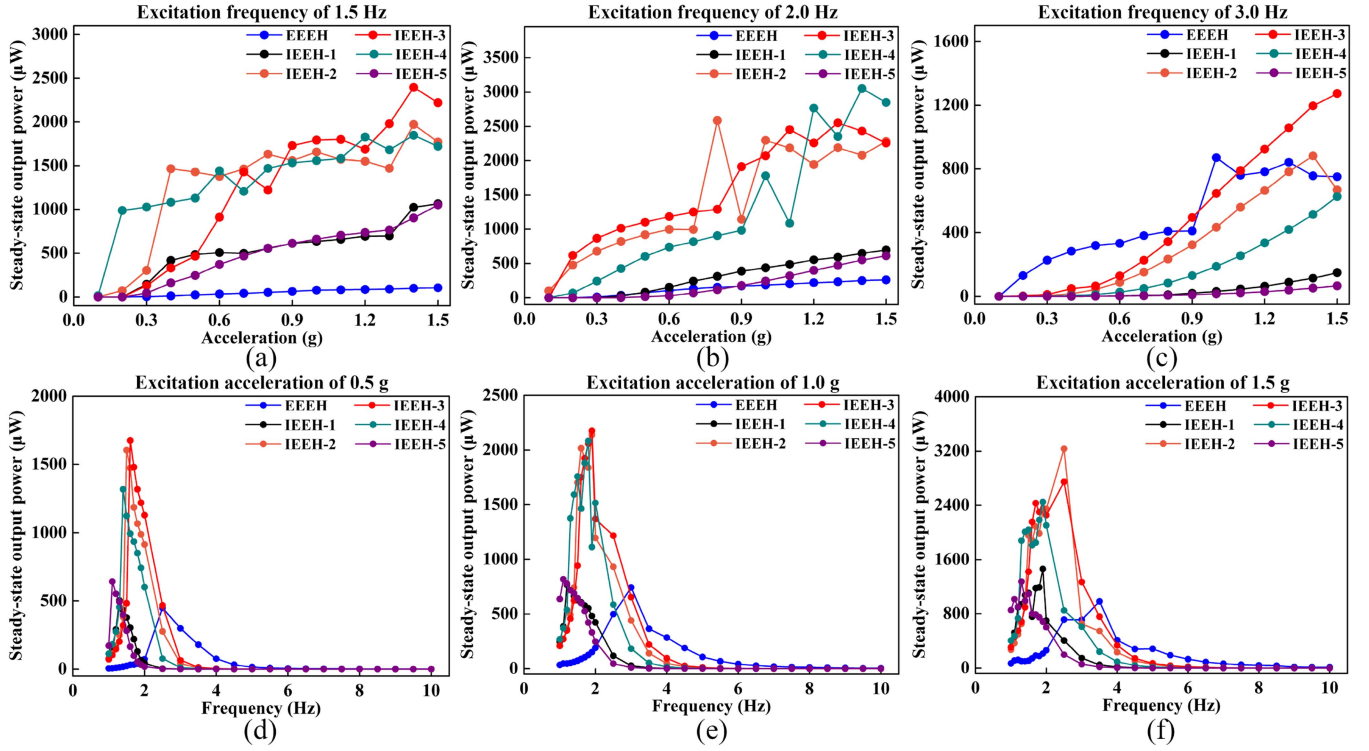


Fig. 8. Power generation performance of the EEEH and the IEEH- $n$  under different excitation conditions. (a)–(c) Steady-state output power versus the excitation acceleration. (d)–(f) Steady-state output power versus the excitation frequency.

### B. Performance Characterization by Bench Tests

$P_{ss}$  is derived by the measured  $V_{oc,rms}$  under different excitation accelerations with a matched load of each harvester. In Fig. 8(a)–(b),  $P_{ss}$  of each harvester prototype generally grows with a larger acceleration. Compared to the EEEH, a several-fold enhancement in  $P_{ss}$  is achieved by the IEEHs due to their enhanced electromechanical transduction. Specifically, the IEEH-2, IEEH-3, and IEEH-4 reach the scale of 1–3 mW with the growing excitation acceleration, while that of the EEEH remains under 0.3 mW. Also,  $P_{ss}$  of these three IEEHs no longer increase smoothly under higher accelerations, e.g., over 0.7 g for the IEEH-3 at 1.5 Hz [see Fig. 8(a)], where the rotor's motion mode changes from reciprocating swing to rotation of a higher intensity. This shifted vibration pick-up performance endows them with higher rotor velocities than the others under such excitation conditions. But the detrimental impact of the added rotor weight by too many DT4C fillers on the vibration pick-up gradually outweighs their benefits in the electromechanical transduction, which accounts for the weaker output performance of the IEEH-5. These results demonstrate the dual impacts of diverse rotor configurations on these two aspects of energy harvesting. Consequently, the IEEH-2, IEEH-3, and IEEH-4 reach a superior  $P_{ss}$  level to the others, for example, over 16.5 times that of the EEEH under 1–1.5 g accelerations at 1.5 Hz.

However, this superiority is diminished at 3 Hz in Fig. 8(c), as the excitation frequency gets closer to the natural frequency of the EEEH (2.87 Hz) instead of those of the IEEHs (1.27–1.97 Hz). Thus, the excitation frequency is also a vital factor

affecting their superior performance. Frequency responses of  $P_{ss}$  by these prototypes are obtained under repeated excitations and presented in Fig. 8(d)–(f). The effective working bands of the IEEHs with significant  $P_{ss}$  levels mainly fall in the ultralow-frequency region below 3 Hz because of their natural frequencies. Taking the IEEH-3 as an example, the generated  $P_{ss}$  reaches 1.68 mW at 1.6 Hz, 2.18 mW at 1.9 Hz, and 2.75 mW at 2.5 Hz under excitation accelerations of 0.5–1.5 g, corresponding to a VPD enhancement of 264.07%, 187.36%, and 174.05% of the maximum  $P_{ss}$  of the EEEH, respectively. Notably, with a larger acceleration, each IEEH- $n$  works effectively with high  $P_{ss}$  in a broader frequency range due to the rotor motion change from swing to rotation, which further expands its output advantages towards the frequencies higher than its natural frequency [see Fig. 8(f)].

The above results are based on the prototypes with fixed inner diameter of the coil sets and wire diameter by the fabrication limits. According to some studies, more coil turns can be achieved by a smaller inner diameter of the coil sets with a fixed outer diameter, but the added MF for a higher EMF does not necessarily make up for the increasing  $R_{in}$  [10]. Likewise, there is a tradeoff for higher  $P_{ss}$  between the increased EMF and added  $R_{in}$  by a smaller wire diameter, due to the nonuniformly distributed magnetic fields shown in Fig. 5 [26]. Also, the fabrication yield rate of fine wire is a practical concern when changing the wire diameter [25]. On this basis, there is still room for further optimization of the IEEHs. To enhance their energy harvesting efficiency by refining all these coil parameters is a complex but feasible engineering problem, as some response

TABLE II  
PERFORMANCE COMPARISON AMONG ECCENTRIC-ROTOR-BASED ENERGY HARVESTERS OF SIMILAR SIZES

| Ref.   | Volume (cm <sup>3</sup> ) | Excitation conditions | Maximum power (mW) | VPD ( $\mu\text{W}/\text{cm}^3$ ) | NVPD ( $\mu\text{W}/(\text{cm}^3 \cdot \text{g}^2)$ ) |
|--------|---------------------------|-----------------------|--------------------|-----------------------------------|-------------------------------------------------------|
| [8]    | 15.93                     | 1 g at 2.5 Hz         | 0.21               | 13.18                             | 13.18                                                 |
|        |                           | 1.5 g at 3.5 Hz       | 0.36               | 22.60                             | 10.04                                                 |
| [12]   | 171.79                    | 0.5 g at 5.6 Hz       | 2.38               | 13.85                             | 55.4                                                  |
| [13]   | 50                        | 0.4 g at 3.45 Hz      | 1.40               | 28                                | 178                                                   |
|        |                           | 0.98 g at 2 Hz        | 1.36               | 27.2                              | 28.32                                                 |
| [16]   | 10.56                     | 1.49 g at 1.3 Hz      | 0.15               | 14.20                             | 6.40                                                  |
| EEEH   | 24.88                     | 0.5 g at 2.5 Hz       | 0.45               | 18.09                             | 72.36                                                 |
|        |                           | 1 g at 3 Hz           | 0.74               | 29.74                             | 29.74                                                 |
|        |                           | 1.5 g at 3.5 Hz       | 0.98               | 39.34                             | 17.48                                                 |
| IEEH-3 | 25.51                     | 0.5 g at 1.6 Hz       | 1.68               | 65.86                             | 263.44                                                |
|        |                           | 1 g at 1.9 Hz         | 2.18               | 85.46                             | 85.46                                                 |
|        |                           | 1.5 g at 2.5 Hz       | 2.75               | 107.81                            | 47.92                                                 |

surface models and evolutionary algorithms have proved useful in such multiparameter optimization issues [10], [27], [28]. Such coil parameter optimization is not the focus of this work, but our future work will include this investigation, taking all these factors into account.

In general, the bench test results have verified the superior performance of our proposed IEEHs to the EEEH, as well as some state-of-the-art eccentric-rotor-based EMEHs of similar sizes, as compared in Table II in terms of the maximum  $P_{ss}$ , VPD, and NVPD versus acceleration for peer comparison [29].

### C. Performance Analysis by Rotor Motion Models

To further comprehend some phenomena in bench tests, detailed analysis is conducted to the rotor movement in the IEEH-3 and the EEEH based on their fitted rotor motion models, as the rotor dynamics are hard to measure directly. MATLAB/Simulink models are built for (11) and (12) before fitting and simulation, and numerically solved by ordinary differential equation (ODE) tools like the ODE3. Repeated sinusoidal signals, identical to those in bench tests, are applied as  $x_{in}(t)$  to these models, while the measured frequency responses of  $V_{oc,rms}$  are used as the target model outputs when  $b_m$  and  $b_e$  are tuned manually. A coefficient of determination ( $R^2$ ) greater than 0.8 serves as the fitting criterion, and the consistency of some representative features between the simulated voltages and the measurements is also considered, particularly the irregularly changing  $V_{oc,rms}$  amplitudes when these rotors start to rotate.

Fig. 9 presents the measured and simulated  $V_{oc,rms}$  of these two harvesters after fitting and shows a good agreement especially at the ultralow frequencies below 5 Hz, which is the focus of our analysis. A closer match at higher frequencies can be possibly achieved using more complex damping models instead of our simplified linear dampers in (10), but at a higher computation cost. The current fitted models are of sufficient confidence for our analysis of the rotor dynamics with negligible errors. Besides, the fitted parameters, i.e., the obtained  $b_e$  of  $1.357 \times 10^{-6} \text{N} \cdot \text{m} \cdot \text{s} \cdot \text{rad}^{-1}$  for the EEEH ( $R^2 = 0.8647$ ) and  $6.958 \times 10^{-6} \text{N} \cdot \text{m} \cdot \text{s} \cdot \text{rad}^{-1}$  for the IEEH-3 ( $R^2 = 0.8144$ ), also validate the enhanced ECC by our approach.

Based on the fitted models, the detailed rotor movement in these two harvesters under some representative excitation conditions are obtained by simulation and shown in Fig. 10,

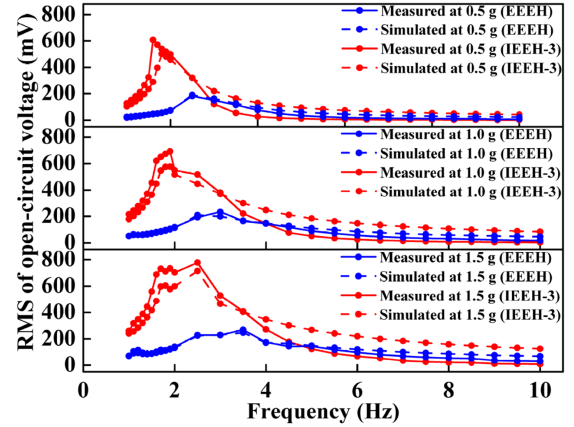


Fig. 9. Fitted results of  $V_{oc,rms}$  measured by the EEEH and the IEEH-3 in bench tests.

along with the corresponding measured  $V_{oc}$  waveforms. Two distinct rotor motion modes are observed for both harvesters, i.e., the reciprocating swing and the rotation. Under the 0.5 g acceleration, the rotors in these harvesters swing steadily in a sinusoidal-like manner as seen in Fig. 10(a). Though the EEEH has higher amplitudes of  $\theta(t)$  and  $\dot{\theta}(t)$  than the IEEH-3 under this excitation in Fig. 10(e) due to its lighter rotor weight (see Table I), the IEEH-3 generates higher  $V_{oc}$  in Fig. 10(i) and  $V_{oc,rms}$  in Fig. 9 because of the enhanced ECC by our designed magnetic rotor. When the applied vibration gets intensified, occasional rotor rotation occurs in the IEEH-3 with a  $\theta(t)$  range over  $360^\circ$ , especially at the excitation frequency of 1.9 Hz near its natural frequency in Fig. 10(b). Its rotor phase trajectory in Fig. 10(f) reveals the accelerated  $\dot{\theta}(t)$  compared to the swing mode in Fig. 10(e), even faster than the EEEH under the 1 g acceleration at 1.9 Hz. This benefits the IEEH-3 in both of its vibration pick-up and electromechanical transduction, so the measured instantaneous  $V_{oc}$  in Fig. 10(j) is greatly amplified. Besides, the change time of its rotor's moving direction evidently decreases in Fig. 10(b) when the rotation happens, which allows its theoretically infinite accelerating stroke to be fully utilized with less deceleration of the moving rotor [3]. Thus, the overall  $\dot{\theta}(t)$  can be kept at a high level in this motion mode.

Moreover, the IEEH-3 outputs a higher  $V_{oc}$  than the EEEH in Fig. 10(k) as well as a greater  $V_{oc,rms}$  in Fig. 9, despite its

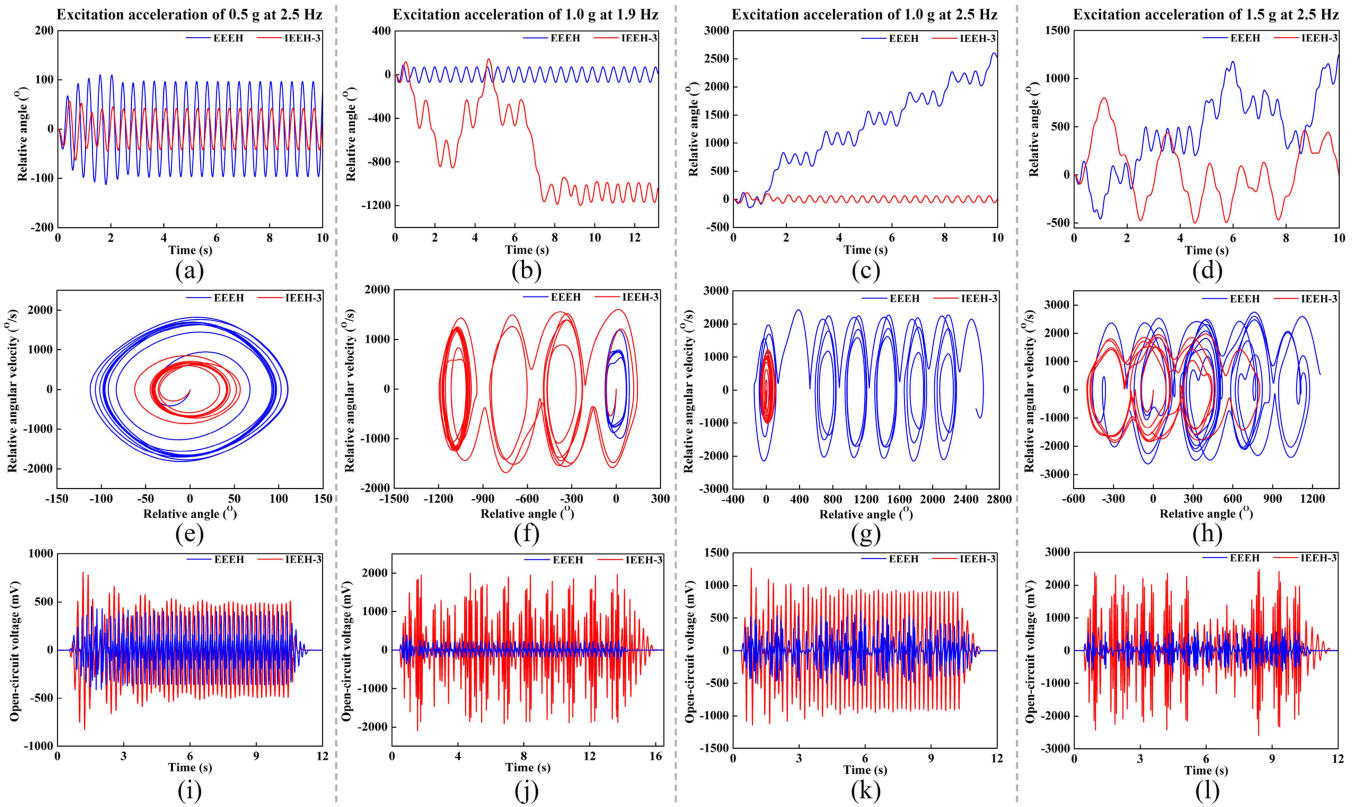


Fig. 10. Simulated rotor dynamics and corresponding measured  $V_{OC}$  waveforms of the EEEH and the IEEH-3. (a)–(d) Simulated time histories of the relative angular displacement. (e)–(h) Simulated phase trajectories of these magnetic rotor motions. (i)–(l) Measured  $V_{OC}$  waveforms.

low-speed rotor swing and the high-speed rotor rotation of the EEEH caused by the excitation frequency of 2.5 Hz near the latter harvester's natural frequency. This output advantage of the IEEH-3 is attributed to its promoted interaction between its CMUs and coils, which outweighs its disadvantage in vibration pick-up under such an excitation. In Fig. 10(d) and (h), when the acceleration continues to rise to 1.5 g, the rotor in the IEEH-3 exhibits continuous rotary motions over a wider band including 2.5 Hz, which leads to a further increased overall  $\dot{\theta}(t)$  and thus a larger  $V_{OC}$  amplitude. The intense rotor dynamics also account for its widened high-output frequency region with a higher optimal working frequency in Fig. 9 when the excitation acceleration grows. This characteristic endows our proposed IEEHs with stronger adaptability to limb motions at volatile frequencies in the ultralow range.

To sum up, the above analysis of the rotor dynamics and the measured  $V_{OC}$  has revealed that the improved VPD and NVPD performance of the IEEHs can be ensured by the synergetic effect of these dual rotor motion modes and the enhanced electromechanical transduction, even if the excitation condition deviates slightly from their natural frequencies.

#### D. Brief Demo of Biomechanical Energy Harvesting

As a demonstration of biomechanical energy harvesting from real limb motions, the prototypes are worn at the wrists and the ankles respectively of some subjects running constantly at

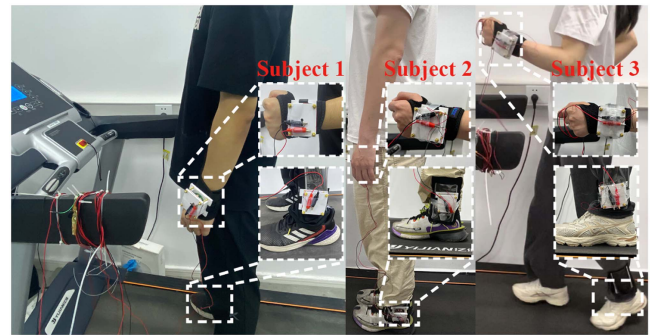


Fig. 11. Photographs of the worn prototypes at the wrists and ankles of different testing subjects.

4 km/h on a treadmill (see Fig. 11). To guarantee the representativeness and credibility of the results, three subjects are involved in our tests (Subject 1: Male/1.71 m/65 kg; Subject 2: Male/1.75 m/69 kg; Subject 3: Female/1.60 m/48 kg), where the same experimental procedure is repeated to keep consistent test conditions. The output voltages ( $V_{out}$ ) are recorded by a source meter unit (Keithley 2450) in each case for the  $P_{ss}$  evaluation with their matched loads. A 3D-printed case with a nylon band is used for ease of prototype wearing.

In all cases the proposed IEEHs obviously outperform the EEEH as shown in Fig. 12(a), with  $P_{ss}$  basically above 1 mW

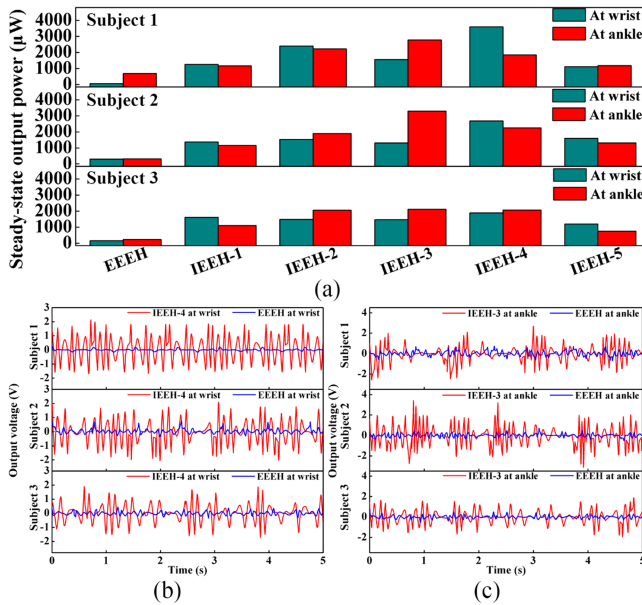


Fig. 12. Comparative results of biomechanical energy harvesting from constantly running subjects at different wearing positions. (a) Steady-state output power of the prototypes. (b) Voltages of the IEEH-4 with the highest power at wrist and the EEEH. (c) Voltages of the IEEH-3 with the highest power at ankle and the EEEH.

both at wrist and ankle, whereas  $P_{ss}$  of the EEEH is not more than  $687.90 \mu\text{W}$ . Also, an agreement in the relative  $P_{ss}$  levels of the prototypes is reached among these subjects, where the IEEH-4 generates the maximum  $P_{ss}$  at wrist (Subject 1:  $3.60 \text{ mW}$ ; Subject 2:  $2.69 \text{ mW}$ ; Subject 3:  $1.89 \text{ mW}$ ) and the IEEH-3 achieves the maximum  $P_{ss}$  at ankle (Subject 1:  $2.78 \text{ mW}$ ; Subject 2:  $3.29 \text{ mW}$ ; Subject 3:  $2.12 \text{ mW}$ ). This attributes to the differentiated adaptability to such limb motions of these IEEHs caused by their diverse rotor dynamics, e.g., the matching between their natural frequencies and different dominant frequencies of the wrist and the ankle [8], [24]. In the above tests, a maximum VPD of  $140.41 \mu\text{W}/\text{cm}^3$  is achieved by the IEEH-4, corresponding to the maximum  $P_{ss}$  of  $3.60 \text{ mW}$  at wrist on Subject 1.

Fig. 12(b) and (c) further present some representative  $V_{\text{out}}$  waveforms of the harvesters with highest  $P_{ss}$  for each subject, i.e., the IEEH-4 at the wrist and the IEEH-3 at the ankle, where increased  $V_{\text{out}}$  amplitudes about  $\pm 2 \text{ V}$  are observed. Taking the IEEH-3 as an example, its superiority in the output voltage level is obvious in terms of both  $V_{\text{oc,rms}}$  ( $0.61\text{--}0.78 \text{ V}$  in bench tests in Fig. 9 and  $0.54\text{--}0.85 \text{ V}$  in these on-body tests for all the subjects) and the maximum instantaneous amplitude (up to  $\pm 2 \text{ V}$  in both bench tests and on-body tests), compared to other studies (e.g., a maximum peak-to-peak voltage of  $1.54 \text{ V}$  with a  $V_{\text{oc,rms}}$  of  $0.12 \text{ V}$  in [30], and a maximum  $V_{\text{oc,rms}}$  of  $0.16 \text{ V}$  in [31]). Thus, it is sure to work with the voltage multiplying circuits for effective low-voltage rectification and energy storage used in these works, like the multistage Villard's voltage multipliers [30], [31]. In view of the instantaneous output levels, the generated voltages are also sufficient to directly cover the voltage thresholds of

the rectifier bridges consisting of some low-forward-voltage diodes, like HSMS-285x series (Avago Technologies) diodes ( $\sim 0.16 \text{ V}$ ) [30], [32]. On this basis, the potential of our proposed IEEHs serving as a sustainable power supply in practical use is revealed.

#### IV. CONCLUSION

To improve the VPD of eccentric-rotor-based EMEHs, we propose a new realization of the magnetic rotor with built-in eccentricity to enlarge the volume proportion of electromagnetic transducing components in a confined harvester volume. By this built-in eccentricity based on various embedded fillers, the interacting area between magnets and coils is widened to promote the electromechanical transduction. The employed ferromagnetic fillers further improve the ECC of our proposed IEEH by improving the spatial MF changing rate, which is verified by both numerical simulation and tests. The impacts of  $t_c$  and diverse CMU configurations in the rotor are thoroughly studied for a higher VPD. Finally, results show that the IEEH-3 with an optimal harvester configuration achieves a maximum NVPD of  $263.44 \mu\text{W}/(\text{cm}^3 \cdot \text{g}^2)$  at  $1.6 \text{ Hz}$ , with an increase of  $264.07\%$  over that by a traditional counterpart in a close volume ( $\sim 25 \text{ cm}^3$ ). Theoretical models of the rotor dynamics and power generation are also built and fitted with sufficient confidence to further analyze the rotor movement underlying these results. Finally, a demonstration of on-body tests on several subjects firmly shows the enhanced VPD of the IEEH, which reveals its promising future to efficiently harvest the biomechanical energy from ultralow-frequency limb motions. In short, our approach provides a new insight for the VPD enhancement of wearable eccentric-rotor-based EMEHs undergoing a strict volume limit.

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#### REFERENCES

- [1] C. Hou et al., "A rotational pendulum based electromagnetic/triboelectric hybrid-generator for ultra-low-frequency vibrations aiming at human motion and blue energy applications," *Nano Energy*, vol. 63, Sep. 2019, Art. no. 103871.
- [2] M. Liu, F. Qian, J. Mi, and L. Zuo, "Biomechanical energy harvesting for wearable and mobile devices: State-of-the-art and future directions," *Appl. Energy*, vol. 321, Sep. 2022, Art. no. 119379.
- [3] Y. Cao et al., "Human-motion adaptability enhancement of wearable electromagnetic vibration energy harvesters toward self-sustained body sensor networks," *Cell Rep. Phys. Sci.*, vol. 5, no. 9, Sep. 2024, Art. no. 102117.
- [4] T. Wang, "Pendulum-based vibration energy harvesting: Mechanisms, transducer integration, and applications," *Energy Conv. Manag.*, vol. 276, Jan. 2023, Art. no. 116469.
- [5] A. S. Farooq and P. Zhang, "A comprehensive review on the prospects of next-generation wearable electronics for individualized health monitoring, assistive robotics, and communication," *Sens. Actuator A-Phys.*, vol. 344, Sep. 2022, Art. no. 113715.
- [6] F. Gao, W.-H. Liao, J. Liang, and X. Wu, "Performance enhancement of piezoelectric bending beam-based human knee energy harvester," *IEEE/ASME Trans. Mechatron.*, vol. 29, no. 4, pp. 2779–2784, Aug. 2024.

- [7] K. Fan, H. Qu, Y. Wu, T. Wen, and F. Wang, "Design and development of a rotational energy harvester for ultralow frequency vibrations and irregular human motions," *Renew. Energy*, vol. 156, pp. 1028–1039, Aug. 2020.
- [8] S. Fan, M. Fu, Y. Zhou, D. Hou, G. Zhang, and Y. Cao, "Ultralow-frequency biomechanical energy scavenging and human activity recognition at different positions using a multifunctional wearable energy harvester," *IEEE Trans. Instrum. Meas.*, vol. 73, May 2024, Art. no. 9510914.
- [9] M. Cai and W.-H. Liao, "A coaxial wrist-worn energy harvester for self-powered Internet of Things sensors," *IEEE Internet Things J.*, vol. 10, no. 1, pp. 133–143, Jan. 2023.
- [10] M. Li, H. Deng, Y. Zhang, K. Li, S. Huang, and X. Liu, "Ultra-low frequency eccentric pendulum-based electromagnetic vibrational energy harvester," *Micromachines*, vol. 11, no. 11, Nov. 2020, Art. no. 1009.
- [11] H. Liu et al., "A non-resonant rotational electromagnetic energy harvester for low-frequency and irregular human motion," *Appl. Phys. Lett.*, vol. 113, no. 20, Nov. 2018, Art. no. 203901.
- [12] W. Wang et al., "Harnessing energy from hand-shaking vibration for electronics through a magnetic rolling pendulum bistable energy harvester," *Energy Conv. Manag.*, vol. 310, Jun. 2024, Art. no. 118466.
- [13] J. Smilek, Z. Hadas, J. Vetiska, and S. Beeby, "Rolling mass energy harvester for very low frequency of input vibrations," *Mech. Syst. Signal Process.*, vol. 125, pp. 215–228, Jun. 2019.
- [14] S. Guo et al., "Mechanism, theory and application research of a rotating electromagnetic energy harvester suitable for multi-directional excitation," *J. Phys. D-Appl. Phys.*, vol. 55, no. 8, Nov. 2021, Art. no. 085503.
- [15] D. Hao et al., "An electromagnetic energy harvester with a half-wave rectification mechanism for military personnel," *Sustain. Energy Technol. Assess.*, vol. 57, Jun. 2023, Art. no. 103184.
- [16] M. Cai and W.-H. Liao, "Enhanced electromagnetic wrist-worn energy harvester using repulsive magnetic spring," *Mech. Syst. Signal Process.*, vol. 150, Mar. 2021, Art. no. 107251.
- [17] H. Zhang, X. Wu, Y. Pan, A. Azam, and Z. Zhang, "A novel vibration energy harvester based on eccentric semicircular rotor for self-powered applications in wildlife monitoring," *Energy Conv. Manag.*, vol. 247, Nov. 2021, Art. no. 114674.
- [18] H. Sharghi and O. Bilgen, "Dynamics of pendulum-based systems under human arm rotational movements," *Mech. Syst. Signal Process.*, vol. 183, Jan. 2023, Art. no. 109630.
- [19] T. Xue, S. Williams, R. Rantz, M. A. Halim, and S. Roundy, "System modeling, characterization, and design considerations for generators in commercial watches with application to energy harvesting for wearables," *IEEE/ASME Trans. Mechatron.* vol. 23, no. 5, pp. 2515–2524, Oct. 2018.
- [20] Y. J. Li et al., "Design and analysis of an electromagnetic turnout for the superconducting maglev system," *Physica C*, vol. 528, pp. 84–89, Sep. 2016.
- [21] Q. Lu, G. Xia, M. Cai, X. Li, J. Cao, and W.-H. Liao, "An ultralow frequency energy harvester with an asymmetric-stiffness pendulum inspired by biological grooming behavior," *IEEE Internet Things J.*, vol. 11, no. 13, pp. 24222–24233, Jul. 2024.
- [22] Y. Li, J. Li, A. Yang, Y. Zhang, B. Jiang, and D. Qiao, "Electromagnetic vibrational energy harvester with microfabricated springs and flexible coils," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2684–2693, Mar. 2021.
- [23] Y. Li, W. Zhang, Y. Zhang, and J. Cao, "A batch-fabricated electromagnetic energy harvester based on flex-rigid structures," *Sens. Actuator A-Phys.*, vol. 263, pp. 571–579, Aug. 2017.
- [24] C. Li, S. Wu, P. C. K. Luk, M. Gu, and Z. Jiao, "Enhanced bandwidth non-linear resonance electromagnetic human motion energy harvester using magnetic springs and ferrofluid," *IEEE/ASME Trans. Mechatron.*, vol. 24, no. 2, pp. 710–717, Apr. 2019.
- [25] M. Cai and W.-H. Liao, "Design, modeling, and experiments of electromagnetic energy harvester embedded in smart watch and wristband as power source," *IEEE/ASME Trans. Mechatron.*, vol. 26, no. 4, pp. 2104–2114, Aug. 2021.
- [26] Y. Peng et al., "Influences of wire diameters on output power in electromagnetic energy harvester," *Int. J. Precis. Eng. Man-GT*, vol. 10, pp. 205–216, Jan. 2023.
- [27] T. N. Phan, J. J. Aranda, B. Oelmann, and S. Bader, "Design optimization and comparison of cylindrical electromagnetic vibration energy harvesters," *Sensors*, vol. 21, no. 23, Nov. 2021, Art. no. 7985.
- [28] L.-D. Liao et al., "A miniaturized electromagnetic generator with planar coils and its energy harvest circuit," *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 4621–4627, Oct. 2009.
- [29] Q. Zhang and E. S. Kim, "Vibration energy harvesting based on magnet and coil arrays for watt-level handheld power source," *Proc. IEEE*, vol. 102, no. 11, pp. 1747–1761, Nov. 2014.
- [30] A. Halim, H. Cho, M. Salauddin, and J. Y. Park, "A miniaturized electromagnetic vibration energy harvester using flux-guided magnet stacks for human-body-induced motion," *Sens. Actuator A-Phys.*, vol. 249, pp. 23–31, Oct. 2016.
- [31] K. Zhang, Y. Su, J. Ding, and Z. Duan, "A low-frequency wideband vibration harvester based on piecewise linear system," *Appl. Phys. A-Mater. Sci. Process.*, vol. 125, no. 5, Apr. 2019, Art. no. 340.
- [32] M. Cai and S. Mirabbasi, "A self-sustained smart monitoring platform for capacitive de-ionization cell in wireless sensor network," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 4164–4172, May 2021.



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